

PROJECT MANAGEMENT

Project Title:

- Health Chatbot

Student Details:

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Supervised By:

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Teaching Assistants:

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Technical Feasibility

The project is technically feasible because it is a small Python-based application. The main idea does not require complicated hardware or a large system. The chatbot can run on a normal computer and its basic logic can be implemented using Python.

Operational Feasibility

The system is simple from the user's point of view. The user selects symptoms from the available choices and receives the related health information.

Economic Feasibility

The project does not require expensive hardware or paid services. The main resources are a computer, Python, and the time needed for development and testing.

Conclusion

The Health Chatbot is feasible as a small academic project.

Project Name: Health Chatbot

Project Objective:

The objective of the project is to create a simple chatbot that provides health information based on the symptoms selected by the user.

Technology:

The project is developed using Python.

Main User:

The main user is a person who wants to select symptoms and see the related information provided by the system.

Main Deliverable:

A working Health Chatbot that accepts symptom selections and displays the corresponding information.

Project Limitation:

The chatbot is an academic project and is not intended to replace professional medical advice.

Project Scope Statement — صفحة 3

In Scope

- Selecting symptoms from the available choices.
- Processing the selected symptoms.
- Displaying the related health information.
- Implementing the main chatbot logic using Python.

Out of Scope

- Online doctor consultation.
- Hospital booking.
- Real-time medical monitoring.
- Replacing a professional medical diagnosis.

Scope Boundary

The project focuses on the interaction between selecting symptoms and receiving the related information from the chatbot.

Health Chatbot Project

1. Planning

- Define project idea.
- Define objectives.
- Define project scope.

2. Design

- Plan the chatbot flow.
- Plan the user interaction.

3. Development

- Write the Python code.
- Implement the chatbot logic.
- Connect symptom selections with the related information.

4. Testing

- Test different selections.
- Check the chatbot responses.
- Fix errors.

5. Documentation

- Prepare the project report.
- Document the project work.

6. Presentation

- Prepare the final presentation.
- Prepare for project discussion.

Activity List — 5 صفحة

Activity	Description
Project Idea	Define the Health Chatbot idea
Requirements	Identify what the system should do
Scope	Define what is included in the project
Design	Plan the chatbot flow
Development	Implement the project using Python
Testing	Test the different symptom selections
Error Fixing	Correct problems found during testing
Documentation	Prepare the project report
Presentation	Prepare the final presentation

The Health Chatbot is a small university project, so its direct cost is relatively low.

Resource	Estimated Cost
Computer/Laptop	0 EGP*
Python	0 EGP
Development Software	0 EGP*
Internet/Electricity	Existing personal expense
Printing/Presentation	Estimated separately

Estimated Direct Project Cost: approximately 0–500 EGP depending on printing and presentation requirements.

*Assuming the student already has access to a computer and development environment.

Note: These are estimated academic project costs, not actual recorded expenses.

Gantt Chart — صفحة 7

Task	Week 1	Week 2	Week 3	Week 4
Project Idea & Requirements	✓			
Scope & Planning	✓			
Design	✓			
Python Development		✓		
Testing			✓	
Error Fixing			✓	
Documentation				✓
Presentation				✓

Project Flow

Planning



Requirements



Design



Python Development



Testing



Error Fixing



Documentation



Final Presentation

The activities follow a logical order. Development starts after the basic planning and design are completed, while testing depends on having a working version of the chatbot.

NETWORK DIAGRAM & CRITICAL PATH

1. Network Diagram Concepts

A network diagram is a graphical representation of the workflow and logical interdependencies between project activities. It provides clear visibility into precedences, enabling project teams to visualize which tasks must finish before subsequent tasks can begin.

2. Project Network Logic

The workflow for the Health Chatbot project follows a well-defined sequential and parallel structure:

- **Node 1 (Start):** Project Initialization & Requirements Specification.
- **Node 2 (Database Setup):** Depends directly on Node 1. Involves SQLite configuration.
- **Node 3 (Chatbot Logic):** Depends directly on Node 2 (requires database connection to store user appointments and logs).
- **Node 4 (Streamlit UI Development):** Can run in parallel with Node 3 or sequentially, integrating frontend components with backend functions.
- **Node 5 (Integration & Final Testing):** Depends on both Node 3 and Node 4 completion.
- **Node 6 (Completion / End):** Final deployment to GitHub and project sign-off.

3. Critical Path Analysis

The critical path is the longest sequence of dependent tasks that determines the shortest possible project duration. For this project, the critical path runs through: **Start → Requirements → Database Setup → Chatbot Logic → UI Integration & Testing → End**, totaling 17 days with zero float/slack time.

COST ESTIMATION

1. Introduction & Approach

Cost estimation is a critical component of project management. For the Health Chatbot project, the cost estimation approach involves breaking down the software development lifecycle (SDLC) into discrete phases: requirements analysis, system design, coding, testing, and deployment. Each phase is evaluated in terms of labor hours, resource requirements, and software licensing or cloud hosting overheads.

2. Detailed Cost Breakdown Structure (CBS)

The total project cost is aggregated using bottom-up estimating techniques across major modules:

- **Development & Programming (Python/Streamlit):** Estimated at 120 hours @ \$25/hr = \$3,000. Covers frontend UI design and backend chatbot logic integration.
- **Database Management & Security (SQLite):** Estimated at 40 hours @ \$30/hr = \$1,200. Covers schema design, user authentication, and data encryption.
- **Quality Assurance & Testing:** Estimated at 30 hours @ \$20/hr = \$600. Includes unit testing, debugging, and user acceptance testing (UAT).
- **Project Management & Documentation:** Estimated at 50 hours @ \$20/hr = \$1,000. Includes risk assessment, EVM tracking, and report generation.

3. Total Estimated Cost Summary

The cumulative baseline cost for the Health Chatbot project totals **\$5,800**. This figure acts as the baseline budget (Planned Value at completion) for subsequent Earned Value Management calculations and tracking.

EARNED VALUE MANAGEMENT (EVM)

1. EVM Overview & Core Metrics

Earned Value Management integrates scope, schedule, and cost measurements to assess project performance and progress objectively. The three foundational metrics are:

- **Planned Value (PV):** The authorized budget planned for the work scheduled to be completed by a given date.
- **Earned Value (EV):** The measure of work actually performed expressed in terms of the budget authorized for that work.
- **Actual Cost (AC):** The total actual cost incurred in accomplishing the work performed up to a specific date.

2. Mathematical Problem & Calculation

Scenario: At the end of Week 4 of the Health Chatbot project, the project management baseline states that the project should have completed 60% of the total work, with a total budget at completion (BAC) of \$6,000. Upon review, actual completion is at 50%, and the total costs incurred so far amount to \$3,800.

Given Data:

- BAC (Budget at Completion) = \$6,000
- Planned % Complete = 60%
- Actual % Complete (EV Work) = 50%
- AC (Actual Cost) = \$3,800

Calculations:

- **PV** = $60\% \times \$6,000 = \$3,600$
- **EV** = $50\% \times \$6,000 = \$3,000$
- **CV (Cost Variance)** = $EV - AC = \$3,000 - \$3,800 = -\$800$ (Over budget)
- **SV (Schedule Variance)** = $EV - PV = \$3,000 - \$3,600 = -\$600$ (Behind schedule)

- **CPI (Cost Performance Index)** = $EV / AC = 3,000 / 3,800 = 0.789$ (Underperforming cost efficiency)
- **SPI (Schedule Performance Index)** = $EV / PV = 3,000 / 3,600 = 0.833$ (Behind schedule schedule efficiency)

Conclusion: Since both CPI and SPI are less than 1.0, corrective actions regarding resource allocation and task acceleration are required.

PERT TECHNIQUE (CONCEPTS & METHODOLOGY)

1. Introduction to PERT

Program Evaluation and Review Technique (PERT) is a statistical tool used in project management designed to analyze and represent the tasks involved in completing a given project, especially treating time uncertainty by employing three time estimates for each activity.

2. The Three Time Estimates

For every project task (such as database configuration, AI model integration, or UI testing), PERT utilizes three distinct time values:

- **Optimistic Time (O):** The minimum possible time required to accomplish a task under ideal, flawless conditions.
- **Most Likely Time (M):** The normal completion time expected under standard working circumstances, accounting for normal operational interruptions.
- **Pessimistic Time (P):** The maximum possible time required if significant delays, technical roadblocks, or critical errors occur.

3. PERT Mathematical Formulas

Expected Time (Te):

$$Te = (O + 4M + P) / 6$$

Activity Variance (σ^2):

$$\sigma^2 = ((P - O) / 6)^2$$

Standard Deviation (σ):

$$\sigma = (P - O) / 6$$

These formulas allow project managers to calculate probabilistic completion durations for critical path activities in software engineering projects.

PERT TECHNIQUE (NUMERICAL PROBLEM)

Detailed Solved Problem for Health Chatbot Modules

Problem Statement: Consider the core development tasks for the Health Chatbot project. The table below lists the Optimistic (O), Most Likely (M), and Pessimistic (P) time estimates (in days) for three critical activities: Task A (Database Setup), Task B (Chatbot Logic Development), and Task C (UI & Testing).

Activity Breakdown:

- **Task A (Database Setup):** O = 2 days, M = 4 days, P = 12 days
- **Task B (Chatbot Logic):** O = 4 days, M = 5 days, P = 12 days
- **Task C (UI & Testing):** O = 3 days, M = 6 days, P = 9 days

Step-by-Step Solution:

1. Calculate Expected Time (Te) for each task:

- **Task A:** $Te = (2 + 4(4) + 12) / 6 = (2 + 16 + 12) / 6 = 30 / 6 = 5.0 \text{ days}$
- **Task B:** $Te = (4 + 4(5) + 12) / 6 = (4 + 20 + 12) / 6 = 36 / 6 = 6.0 \text{ days}$
- **Task C:** $Te = (3 + 4(6) + 9) / 6 = (3 + 24 + 9) / 6 = 36 / 6 = 6.0 \text{ days}$

2. Calculate Total Project Expected Duration:

Assuming sequential dependency along the critical path (A → B → C):

$$\text{Total Expected Time} = Te(A) + Te(B) + Te(C) = 5.0 + 6.0 + 6.0 = 17.0 \text{ days.}$$

9. PERT

PERT (Program Evaluation and Review Technique) is used to estimate the expected time needed to complete a project activity when the exact time is uncertain.

Time Estimates

- **Optimistic (O):** Shortest expected time.
- **Most Likely (M):** Normal expected time.
- **Pessimistic (P):** Longest expected time.

Formula

$$\text{Expected Time} = (O + 4M + P) / 6$$

Example

$$O = 2 \text{ days}$$

$$M = 3 \text{ days}$$

$$P = 5 \text{ days}$$

$$\text{Expected Time} = (2 + 4 \times 3 + 5) / 6 = 3.33 \text{ days}$$

PERT can be used in the Health Chatbot project to estimate the time needed for development, testing, and documentation.

CPM & Critical Path — 10 صفحة

CPM is used to identify the activities that can affect the total project duration.

For the Health Chatbot, the main sequence can be:

**Planning → Requirements → Design →
Development → Testing → Error Fixing →
Documentation → Presentation**

These activities are important because a major delay in one of them can affect the completion of the project.

The exact critical path depends on the final durations assigned to each activity.

Earned Value Management (EVM) — صفحة 11

EVM is a project management method used to compare planned progress with actual progress.

Main Terms:

- **PV (Planned Value):** The value of the work planned to be completed.
- **EV (Earned Value):** The value of the work actually completed.
- **AC (Actual Cost):** The actual cost spent.

Main Indicators:

$$CPI = EV / AC$$

$$SPI = EV / PV$$

For this small academic project, EVM can mainly be used to monitor whether the project is progressing according to the planned schedule and estimated resources.

Risk Analysis — 12 صفحة

Risk	Impact	Response
Coding Errors	Medium	Test the chatbot and fix errors
Time Delay	High	Divide the project into smaller tasks
Interface Problems	Medium	Test the interface and simplify it
Incorrect Information	High	Review the information before finalizing
Presentation Problems	Medium	Practice the project explanation

Risk Management Approach:

The main risks should be identified early and checked during development and testing.

The Health Chatbot follows a simple process.

Step 1: The user starts the chatbot.

Step 2: The available symptoms are presented to the user.

Step 3: The user selects the symptoms that match the situation.

Step 4: The Python program processes the selected symptoms.

Step 5: The chatbot displays the related health information.

Main Flow:

Start → Select Symptoms → Process Selection → Display Information

The system is designed to provide information based on the available project data.

Human-Computer Interaction (HCI) — 14 صفحة

HCI focuses on the way the user interacts with the Health Chatbot.

The interaction is kept simple because the main user action is selecting symptoms and receiving information.

The system should make it clear:

- What the user can select.
- What happens after selecting symptoms.
- Where the related information appears.

The main goal is to make the interaction understandable without requiring complicated steps from the user.

The interface is designed around the main function of the project.

Clear Choices:

Symptoms should be presented as understandable choices.

Simple Interaction:

The user should not need many steps to get the related information.

Readable Information:

The response should be displayed in a clear and readable way.

User Friendliness:

The simple interaction makes the system easier for a user to understand and use.

The interface should focus on the main purpose instead of adding unnecessary elements.

Code Quality, Complexity & — 16 صفحة Maintainability

Programming Language:

The project is implemented using **Python**.

Code Quality:

The code should be organized and use understandable names and simple logic.

Complexity:

The project has a relatively small scope. The main processing depends on the symptoms selected by the user and the corresponding information.

Maintainability:

The chatbot can be maintained by updating the available symptom information or modifying the chatbot logic when needed.

Testing:

Testing the different possible selections helps find errors and improves the reliability of the final version.

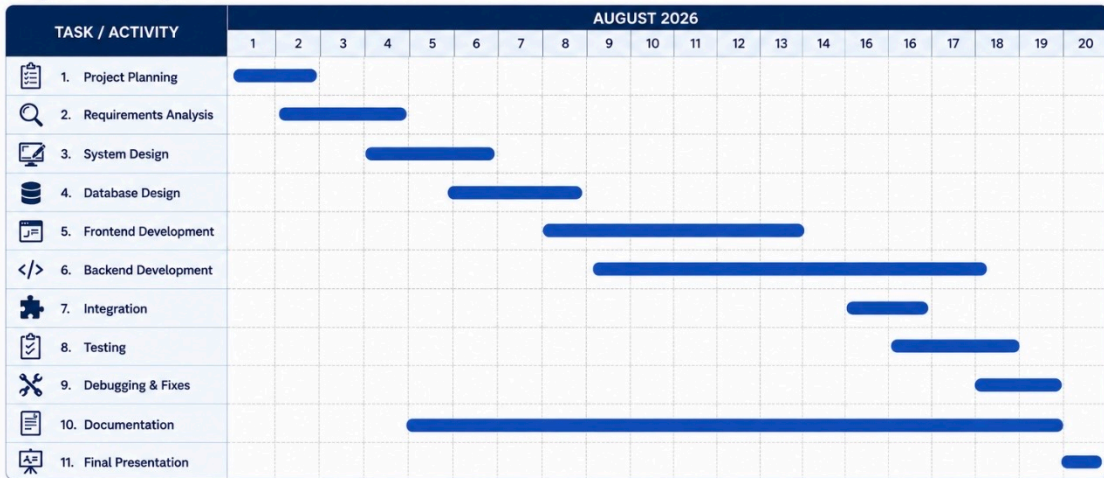


GANTT CHART

Health Chatbot Project



Project Start: 1 August 2026
Project End: 20 August 2026



 = Task Duration



Project Summary

This Gantt chart shows the schedule of tasks for the Health Chatbot project from the start date (1 August 2026) to the end date (20 August 2026).



Goal

To build an intelligent and user-friendly health assistant chatbot.

GANTT CHART

1. Overview and Purpose

A Gantt chart is a horizontal bar chart developed as a production control tool that illustrates a project schedule. In the context of the Health Chatbot project, the Gantt chart maps out project tasks against calendar timelines, explicitly showing start dates, duration, completion milestones, and task overlaps.

2. Task Scheduling Breakdown

The timeline distribution across the 17-day development cycle is structured as follows:

- **Requirements & System Design (Days 1–3):** Gathering clinical user stories, designing database ER diagrams, and setting up repository architecture.
- **Database Implementation (Days 4–8):** Creating SQLite tables (`healthcare.db`), writing Python integration scripts, and testing user accounts (`users.json`).
- **Chatbot AI Logic & Streamlit UI (Days 9–14):** Implementing symptom triage logic, building interactive web interface pages via Streamlit, and session state management.
- **Testing, Debugging & Deployment (Days 15–17):** Conducting end-to-end testing, refining exception handling, pushing the repository to GitHub, and preparing final documentation.

3. Milestone Tracking

Milestone 1 is achieved upon database readiness (Day 8); Milestone 2 is achieved upon core chatbot integration (Day 14); Final Project Sign-off occurs at Day 17.

THANK YOU

Questions?

Health Chatbot

AI-Based Health Assistance System

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Project Management